

- 9 (a) nucleus emits α -particles or β -particles and/or γ -radiation
to form a different / more stable nucleus B1
B1 [2]
- (b) (i) fluctuations in count rate (not 'count rate is not constant') B1 [1]
- (ii) no effect B1 [1]
- (iii) if the source is an α -emitter B1
either α -particles stopped within source (and gain electrons)
or α -particles are helium nuclei B1 [2]
- allow 1/2 for 'parent nucleus gives off radiation to form daughter nucleus'